

Quotient Factorization.

Thm. Let $\varphi: G \rightarrow H$ be a group homomorphism, and $N \trianglelefteq G$ be a normal. Then, $N \subseteq \ker \varphi$ if and only if there exists a unique homomorphism

$$\overline{\varphi}: G/N \rightarrow H \text{ such that } \varphi = \overline{\varphi} \circ \pi$$

where the $\pi: G \rightarrow G/N, g \mapsto gN$ is the Canonical Projection.

Proof. Suppose that $N \subseteq \ker \varphi$, define

$$\overline{\varphi}: G/N \rightarrow H, gN \mapsto \varphi(g).$$

It is well-defined, because

$$g_1 N = g_2 N \Leftrightarrow g_1^{-1} g_2 \in N \Rightarrow g_1^{-1} g_2 \in \ker \varphi \Leftrightarrow \varphi(g_1^{-1} g_2) = e \Leftrightarrow \varphi(g_1) = \varphi(g_2)$$

And, since φ preserves the operation, so $\overline{\varphi}$ is Homomorphism.

And, given $g \in G$, $(\overline{\varphi} \circ \pi)(g) = \overline{\varphi}(gN) = \varphi(g)$, so $\overline{\varphi} \circ \pi = \varphi$.

To show the Uniqueness, let $\psi: G/N \rightarrow H$ be a map satisfying $\psi \circ \pi = \varphi$.

Given $gN \in G/N$, $\psi(gN) = \psi(\pi(g)) = (\psi \circ \pi)(g) = \varphi(g) = \varphi(gN)$,
therefore $\psi = \overline{\varphi}$.